

Em Peplinski

Early-Career Astrophysicist, Planet Enthusiast

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[he/him/his](#)

Education

- August 2025 - **Ph.D. in Astronomy**, *University of Maryland*, College Park, MD
Present
- August 2020 – **B.S. in Astronomy & Astrophysics with Honors**, *University of Michigan*,
May 2024 Ann Arbor, MI

Research Experience

- February 2026 - **Graduate Research Assistant**, *University of Maryland*, College Park, MD
Present Advised by Professor Megan Weiner Mansfield.
- September 2022 **Undergraduate Research Assistant**, *University of Michigan*, Ann Arbor, MI
- August 2025 Advised by Professor Emily Rauscher and graduate student Thomas Kennedy.

Publications

E. E. Peplinski & Thomas Kennedy, Emily Rauscher, Eliza M.-R. Kempton, and Arjun B. Savel. Limb effects meaningfully impact hot Jupiter emission spectra at JWST precision. (*in revision*).

Talks

- October 2023 **“The Impact of Geometry on Spectra Calculated from 3D GCMs”**,
Great Lakes Exoplanet Area Meeting, Bloomington, IN

Poster Presentations

- April 2024 **“The Impact of Geometry on Spectra Calculated from 3D GCMs”**,
University of Michigan Astronomy Undergraduate Research Symposium, Ann Arbor, MI
- January 2024 **“The Impact of Geometry on Spectra Calculated from 3D GCMs”**,
Conference for Undergraduate Women in Physics, Ann Arbor, MI
- April 2023 **“Creating JWST Emission Spectra from 3D Exoplanet Models”**, *University of Michigan Astronomy Undergraduate Research Symposium*, Ann Arbor, MI

Grants, Scholarships, and Fellowships

- September 2023 **H. Richard & Denise Elmquist MMB Family Scholarship**, \$4,650 USD,
University of Michigan
Merit-based scholarship for members of the Michigan Marching Band (MMB).
- May 2022 **Polish Language Studies Scholarship**, \$750 USD, University of Michigan
To be awarded, students must submit an application, take two consecutive terms of the appropriate Polish language sequence, and receive a B- or better in both courses.

Honors and Awards

- April 2023 **Inducted into the Phi Beta Kappa Honor Society**, *University of Michigan*
Nomination is based on strength of academic achievement and is most commonly awarded to graduating seniors. Was one of thirty-five juniors at the University of Michigan nominated in 2023.
- April 2022 **Annual Award for Excellence in Polish Language Studies**, *University of Michigan*, \$250 USD
Faculty-nominated award recognizing student work that stands out from their peers and embodies creativity and an understanding beyond the classroom.
- 2021 - 2024 **3x James B. Angell Scholar Designation**, *University of Michigan*
Awarded to students who have all grades of A+, A, or A- for two or more consecutive terms.
- 2020 - 2024 **7x University Honors Designation**, *University of Michigan*
Awarded to students who earned a 3.5 grade point average or higher during a term.
- March 2021 **William J. Branstrom Freshman Prize**, *University of Michigan*
Awarded to first-term freshmen who rank in the top 5% of their class.

Teaching and Outreach

- September 2025 - Present **Teaching Assistant for ASTR101**, *University of Maryland*, College Park, MD
Responsible for preparing and teaching a weekly discussion to reinforce lecture material, running two labs each week, and attending lectures to monitor and answer student questions. Other duties include hosting a weekly office hour, proctoring exams, and grading student coursework.
- September 2023 - April 2024 **Telescope Operator**, *University of Michigan*, Part-Time, Ann Arbor, MI
Learned how to operate the 0.4m Ritchey-Chretien reflector telescope in the Angell Hall Observatory at the University of Michigan, along with several smaller telescopes. With this training, helped run weekly observing nights for introductory astronomy courses and answer questions about the telescopes as well as general astronomical knowledge. Additionally helped run observing open houses for the Student Astronomical Society.

Organization Membership

- August 2020 - April 2024 **Michigan Marching Band**, *University of Michigan*, Ann Arbor, MI
Joined the **student leadership team as a Rank Leader** in August 2023, helping to instruct marching technique, refine section performance, and act as a liaison between the directors and section members. Student leadership is also required to attend weekly leadership meetings and host marching office hours. The ensemble rehearses Monday-Thursday for 1.5 hours a day during the Fall semester. During weeks with home games (or away games against rivalry teams), there is an additional 2 hour rehearsal on Friday and a full-day commitment on Saturday. When applicable, post-season travel is further required.
- August 2020 - April 2024 **Student Astronomical Society**, *University of Michigan*, Ann Arbor, MI
Assisted with running the Angell Hall Observatory for public open houses, including operating the telescope when the weather was clear or giving tours when it was cloudy.

Skills

- Programming Experienced – Python, C/C++, LaTeX
Intermediate – MATLAB

Beginner – CSS, HTML
Language English – First Language
Polish – Conversational